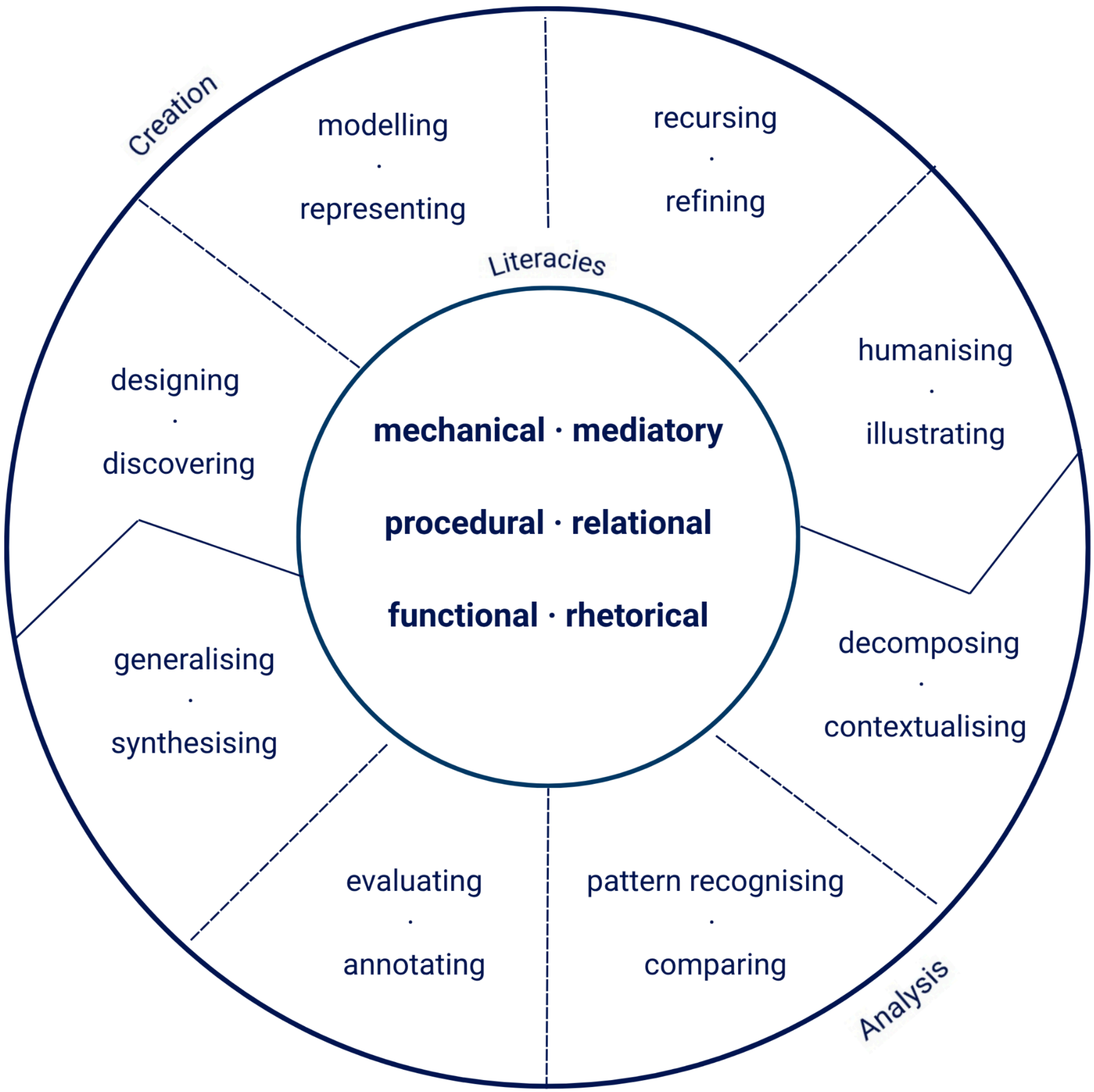
A Competency Framework for **Digital · Humanities** Computing



Disposition: Humanistic Computing

- **Human History, Human Future:** Humanistic computing should be a practice in which computing is both situated in the long history of the human experience and continually used to help understand what it means to be human. This means situating computing in a long history of media and technology and not seeing technology only as an influence on society, but a product of it. Looking forward, it means interrogating the impact of society becoming increasingly digitised on human identity.

- **Plurality and Complexity:** The power of computation is drawn from its facility for information processing at a speed and scale beyond human capability. Prerequisite to this is normalisation and abstraction that would seem to stand in opposition to the plurality and complexity valued in humanistic inquiry, which is always interested in seeing things from a multiplicity of perspectives. Humanistic computing should be about questioning and reorienting the priorities of digital systems to value and preserve plurality, complexity, ambiguity, and variability.

- **Socially-Oriented and Ethically-Focused:** Humanistic computing should place a focus on the social, ethical, political and ecological dimensions of computing. This entails a constantly critical practice and a deliberate choice of application areas, but also, crucially, the integration of humanistic theories and frameworks, in part to help interrogate and counter a history of power and privilege.

Designing · Discovering

Learners are able to:
-identify the questions they would like to pose of a resource or that they are looking to address with a proposed tool
-describe how they might answer these questions using different digital representations, tools, or methods, and compare possible approaches

Mechanical · Mediatory: Learners are able to
- identify and describe the types of tools/technologies needed and approximate hardware and infrastructural requirements of potential paths
-explain how the affordances of these technologies make them similar to and set them apart for existing analog methods

Procedural · Relational: Learners are able to give a high-level description of intended outcome and the network of steps required to reach it.

Functional · Rhetorical: Learners are able to justify/refute technical approaches by linking their affordances to research goals and contextually appropriate theory.

Modelling · Representing

Learners can create a ‘datafication’ of a physical source (a process which may include both digitisation and encoding) or create a computational representation of a problem or method/process.

Mechanical · Mediatory: Learners can
-understand how information is represented/stored by the machine and, at a higher level, by specific digital tools
-explain how and why they have adapted/transformed their source/question for computational constraints/affordances.

Procedural · Relational: Learners are able to specify/describe
-the mapping between problem/source elements and components of a digital model, including how the components are connected/interdependent.
-the relationship between data and context, including questions of consent and sovereignty

Functional · Rhetorical: Learners can explain the implications/justify of modelling choices (e.g. what dimensions were included/excluded, what transformations occurred, how relationships are represented) in relation to their base questions/problem and theoretical lens(es)

Recursing · Refining

Learners are able to:
-incrementally construct digital components for problem solving and interpretation that build upon their own work or that of others
-modify existing methods and tools for their own context and explain the changes in terms of concrete outcome and theoretical grounding
-situate components and identify their effect in the context of larger systems — technical and social alike

Mechanical · Mediatory: Learners are able to
-render imagined processes programmatically
-identify places where this representation differs from the initial conceptualisation
-engage with questions of versioning and compatibility

Procedural · Relational: Learners have an understanding of computational building blocks and pipelines; they can describe the relationships between them in terms of inputs/outputs and dependencies, including interactions with humans.

Functional · Rhetorical: Learners can read/interpret an assemblage of components/processes in terms of the argument(s) they are collectively making.

Humanising · Illustrating

Learners are able to:
- describe the implications of their work — the new context-specific knowledge it represents as well as how it fits into larger networks of meaning-making
-describe their work in terms of the human labour and environmental resources it used and may continue to require

Mechanical · Mediatory: Learners know how to
-document their technical processes for consumption and reuse by others
-identify the technical and labour requirements awareness of long term upkeep/preservation.

Procedural · Relational: Learners can interpret each element of their process, explaining their purpose and limitations with respect to knowledge production, including their relationship to relevant methodologies and theories.

Functional · Rhetorical: Learners are able to
-communicate the meaning/contribution of their work to both technical and problem-domain experts, as well as the lay public
-explain the interplay of computation and human interpretation in a digital object.

Decomposing · Contextualising

Learners are able to build an understanding of objects of interest by breaking them down into component parts and learning about the domain, theoretical and technical context of these pieces and the object as a whole.

Mechanical · Mediatory: Learners are able to:
-parse any available documentation related to the technical composition of digital objects of study or, in its absence, propose possibilities based on its behaviour and context of creation
-interrogate the origins of the technologies and how this has shaped their functionality/use

Procedural · Relational: Learners are able to
-break down objects into a series of linked procedures and make inferences about both the purpose and contextual/theoretical implications of these components and their relationships
-draw on examples from related contexts to identify the dimensions or types of interactions that are not supported

Functional · Rhetorical: Learners are able to propose a potential reading of the context (cultural, temporal), assumptions (ideological, practical) and positionality underpinning the object/tool through examining its mechanics and procedures and how these fit (or not) with existing standards/conventions and current ideas.

Pattern Recognising · Comparing

Learners are able to extract high-level patterns from specific implementations, methodologies, and contextually anchored problems so as to be able to put them in conversation with different but related currents of work

Mechanical · Mediatory: Learners are able to
-abstract upward to describe implementation in terms of types and technologies and patterns of use
-consider these alongside other instances of use to describe and analyse differences across contexts

Procedural · Relational: Learners are able to
-identify the use of common methods (e.g. topic modelling, stylometric analysis) and cross-cutting technical standards (e.g. encoding schemas like TEI, ontology vocabularies like dublin core) and analyse how and why they are used, with respect to other objects/projects that also leverage them
-identify and describe the modes of interaction and knowledge creation these procedures support, interrogating their epistemological foundations

Functional · Rhetorical: Learners are able to situate the project/object in the larger scholarly landscapes, as a function of the technologies it uses, methods it employs, and contextual purpose and arguments and discuss different currents of relevant DH work in terms of these technical or contextual dimensions.

Evaluating · Annotating

Learners are able to offer their own interpretation of the purpose and utility of a computational object and argue the extent to which it is serving its apparent goals using evidence tied to specific elements, behaviours, or examples.

Mechanical · Mediatory: Learners have the capacity to read and interpret an implementation to examine the influence of computational mediation on knowledge representation/claims and how the specificity/affordances of the digital medium are reflected in/leveraged by the object

Procedural · Relational: Learners can propose metrics and methods by which different components of a system and the system as a whole might be tested or validated that are founded in the domain and theoretical context

Functional · Rhetorical: Learners are able to analyse and critique the approach and knowledge contribution of an object through different theoretical lenses/paradigms.

Generalising · Synthesising

Learners are able to apply their contextual and technical learning by integrating it with other work, and view it on an abstract level, thinking in terms of high-level practices and principles.

Mechanical · Mediatory: Learners are able to specify how computational problem solving components might be made more polymorphic/generally usable and discuss the pros and cons of doing so.

Procedural · Relational: Learners are able to describe possible uses (and misuses) for the different components of an object, or the object as a whole, outside of the current context

Functional · Rhetorical: Learners are able to use (or propose a potential use of) the function/argument of an object as part of the investigation of a broader problem. They are also able to discuss its an object as a cultural and political object, identifying its relationship to the cultural context from which it emerged and identifying the groups and systems it has the potential to impact (positively or negatively)